

William Scott Paka

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Summary

- **Senior Data Scientist** with 8 years of experience in natural language processing, machine learning, deep learning, recommendation engines and computer vision.
- **M.Tech in AI from IIT Delhi**
- Published **5 papers** and holding **5 patent** building state-of-the-art models in **NLP**.

Experience

Walmart

Senior Data Scientist | 10/2024 - Present

- **Developed and launched** the "Overall Pick" badge for the Canadian market, driving a **\$20M lift** in GMV and a **+3 bps** increase in conversion.
- **Optimized** the SmartSubs system for the Canadian market, achieving **\$3M in cost savings** (a **44% reduction**) while increasing overall coverage to **60.3%**.
- **Designed and deployed** D40, a SmartSubs extension tailored for medical domains, which streamlined the data pipeline and significantly reduced reliance on external agencies.
- **Built** the "Customer Choice Pick" feature in Canada, to enable users to seamlessly select replacement products for out-of-stock (OOS) items.
- **3 Filed & 1 Approved Patents** on various ideas in collaboration with team.
- **Owning** the development on features for the Mexican and Canada market, including AuthHold, the "Low in Stock" badge, and "Overall Pick" badge.

PromptCloud Technologies

Senior Data Scientist | 12/2021 - 10/2024

- Worked on e-commerce **Products Matching** using textual data along with image features extracted from traditional and pretrained models with an **accuracy of 94%**.
- Trained Vicuna and Mistral **LLM (GenAI)** models to perform classification and generative tasks with an accuracy of 93%.
- Built Category Matching **hierarchical model** which matches products to their standardised categories in hierarchies with an **accuracy of 87%**.
- Built ONET job classification model which has over **1000 classes** with an accuracy of 89%.
- Built a model to identify brand logos on screenshots using **template matching** method, where the sizes vary with accuracy 99%.

KaHa Technologies (Cove IoT)

Senior Data Scientist | 08/2021 - 12/2021

- Deployed **Drowsiness Detection model** using PPG signals with limited and noisy data with an accuracy of 93%.
- Built a hybrid approach to Non-Invasive Continuous Blood Pressure Estimation using PPG signals.

Tata Consultancy Service Research (TRDDC / TCS Research)

Deep Learning and Artificial Intelligence Researcher | 09/2020 - 08/2021

- Built a **persuasive** chatbot system on **RL-based multi-agent** conversations with common-sense reasoning, personality and causality.
- Worked on **code translation and summarization** using a **pivot language** without parallel corpora.
- Worked on **facial image biometrics** for macaques using siamese network.

Covenant IT Solutions

Data Scientist | 03/2016 - 06/2018

- Led development of ML models for Product Recommendations and user review analysis, integrating them into the platform.
- Collaborated with marketing & project management on new initiatives in Data Science that aid in better user experience.

Programming

Python, Java, C, C++

Technology / Utilities

Python, PySpark, PyTorch, Tensorflow, Keras, OpenCV, NLTK, MongoDB, Sklearn, AWS, Data Analysis, NLP, Deep Learning, Machine Learning, Flask, SQL, Statistics, Time Series, Spark/Hadoop, Seaborn/Matplotlib, Git, Jupyter, Spacy, Docker, LLM, GenAI

Skills

NLP, ML, DL, MLOPs, Data Analysis, Computer Vision, OOPs, Data Structures & Algorithms, Problem Solving.

Education

IIT | Delhi, India

M.Tech. - Artificial Intelligence | 08/2020

CGPA 8.45/10

Karunya University | Coimbatore, India

B.Tech. - Computer Science and Engineering (CSE) | 08/2016

CGPA 8.2/10

Patent and Paper Publications

- Paper accepted in PAKDD'21 conference proposed ENDEMIC, a semi-supervised co-attention network which can early detect COVID-19 fake news with an accuracy of 97%. ([PAKDD 2021](#))
- Paper Accepted in ELSEVIER'21, proposed Cross-SEAN, a Cross-Stitch Semi-Supervised Neural Attention Model for COVID-19 Fake News Detection with an accuracy of 95%. ([ELSEVIER 2021](#))
- Masters Thesis on embedding-based joint-sentiment model for short text showing an average improvement of 10% in topic coherence and 5% in topic diversification. ([ICWSM 2021](#))
- Built Blackmarket tweet detection model classifying source of tweets (from blackmarket or genuine) with F1-score of 0.89. ([ASONAM 2019](#))
- Conducted conference on multi modal data - textual, visual & audio with 50+ participants. ([SemEval-2020](#))

Internships | Data Science

Leibniz University, Germany | 01/2020 - 06/2020

- Developed ML models to **Predict Dementia risk** with sparse, high-dimensional data and designed an **automata-based Graph simulation model** for COVID-19 spread analysis and forecasting.

UnitedHealth Group (Optum Global Services) | 04/2019 - 08/2019

- Developed and deployed an end-to-end LDA-based model to automate detection of fraudulent Medical Discharge Summaries, **presented at LIC Delhi and currently in use by NHA**. Enhanced LDA for short texts by **5-8%** through sentiment analysis.

Certifications

- Machine Learning - [Stanford University](#)
- Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization - [DeepLearning.AI](#)
- Neural Networks and Deep Learning - [DeepLearning.AI](#)
- Structuring Machine Learning Projects - [DeepLearning.AI](#)

Achievements

- Teaching Assistant of course **Machine Learning for PG** at IIITD 2019.
- **Ranked 14**, in Hard level Hacker Rank Machine Learning Challenge – [Guess the Flipkart](#)
- **2nd prize** in ‘King Coders’ (2016), a coding Event Conducted by ‘Computer Society of India’
- **2nd prize** in COD++ (2015), a coding Event at a National Level Technical Symposium ‘Expertise 2k15’